for quality assurance, and the datasets we introduce similarly have some process for quality assurance, we note no unified standard has been set to ensure all datasets are sufficiently valid. Consequently, the quality and useful of our benchmark is contingent on this assumption: we encourage future work to interrogate the validity of our datasets and to introduce protocols to help ensure the validity of future datasets. Since benchmarks incentivize future work to build models that improve on the benchmark, validity is especially critical in light of the oft-cited Strathern's Law (Strathern, 1997) — "When a measure becomes a target, it ceases to be a good measure" (also see Goodhart, 1984; Linzen, 2020; Bowman & Dahl, 2021).

Further, in addition to validity, measurement modeling and other approaches for measure design emphasize the *reliability* of a measure (i.e. the measure should not be overly sensitive to undesired sources of variation such as properties of the specific annotators who labelled data). Akin to validity, we lack uniform evidence for the reliability of our measurement. In particular, we highlight the importance of significance testing for making meaningful comparisons given we only run 3 random seeds (due to cost) for selecting in-context examples and we evaluate on 1000 instances instead of the full validation/test set for a given dataset. That is, both of these settings for our evaluation may render some of our claims statistically insignificant; we encourage future work to consider how to better address significance given the scale of this evaluation.[85]

11.3 Limitations of HELM design

Given the nature of our benchmark design, we highlight the important question of aggregation (see Ethayarajh & Jurafsky, 2020; Ma et al., 2021). In comparison to prior work, where a model would receive a single score (e.g. SQuAD F1 accuracy) or a score vector (e.g. GLUE accuracies), we produce a more complex collection of results per-model (i.e. a score matrix of scenarios $\times$ metrics). We believe this is necessary to capture the complexity of the artifacts we characterize (Liao et al., 2021): the generality of language models and the plurality of desiderata we should require of such systems.[86]

However, this complexity does come at a serious cost. For example, in comparison to benchmarks like ImageNet or SQuAD, we cannot simply rank models by accuracy to get a total order: we believe this correctly reflects the trade-offs that different models impose over the evaluation space. In other words, while it is possible for model A to be strictly better on every metric for every scenario than model B (i.e. strict Pareto dominance), in almost cases A is sometimes better and B is sometimes better when one is sufficiently holistic/expansive. To designate A or B as better requires making a judgment that (implicitly/explicitly) weighs the circumstances in which each is better than the other.

Practically, by significantly increasing the volume of results we report for each model, our benchmark could overload a consumer of the results, making it difficult to interpret or act upon them. By providing structure (e.g. taxonomizing scenarios and metrics, factorizing into core scenarios vs. specific components), we hope to retain clarity as we provide nuance. Overall, the detail of our benchmark exposes decision points for different stakeholders to prefer one model over the other based on their values, preferences, and circumstances (e.g. an organization deploying a model on mobile should assign higher priority to the efficiency results).

We leave the question of aggregating model performance to a single number (or a single number per scenario, or per metric) to future work. We do not believe there exists a universal aggregation that satisfies all preferences, reflects all values, or captures all circumstances appropriately. However, we do believe single-number metrics, though reductive, are useful practical tools to simplify decision-making. [87]

[85] We do note that we perform paired bootstrap tests for all model comparisons made in the human evaluations (§8.5: HUMAN-EVALUATIONS).

[86] We this said, we note that using our holistic evaluation as an oracle of sorts, one then may be able to identify minimal evaluations that discriminate between different models. In other words, holistic, broad, and expansive evaluations like HELM complement and possibly could inform succinct unit-testing approaches like HANS (McCoy et al., 2019), CheckList (Ribeiro et al., 2020) and LMentry (Efrat et al., 2022).

[87] By analogy, single-number metrics like GDP and calories are very helpful in practice, albeit flawed and reductive in encoding the true complexities of the economy and nutrition.

12 Conclusion

Language models have transformed AI, ushering in the paradigm of foundation models. The reach of modern language models extends well beyond research, with language models being rapidly productionized into consequential and ubiquitous language technologies, which we expect to only increase in the near-term future. We lack transparency on language models at present, which is especially concerning given their rapid growth and burgeoning impact: as a community, we do not understand language models in their totality. For this reason, we have pushed for holistic evaluation in this effort, as we believe holistic evaluation is a critical means for providing the necessary transparency for language models.

Transparency begets trust and standards. Viewing benchmarks as models for social change, given they orient the development of AI systems, our broader objective is to transform foundation models from immature emerging technologies to reliable tools that support human flourishing. With this objective in mind, we recognize the history and trajectory of AI benchmarking aligns with institutional privilege (Koch et al., 2021). Benchmarks set the agenda and orient progress: we should aspire for holistic, pluralistic and democratic benchmarks (Birhane et al., 2022). Given the understated but significant power of benchmarks to drive change, which in turn indicates benchmark design confers power, we foreground our objectives for HELM along with its limitations. We hope the community will interrogate, adopt, and improve HELM going forward to actualize the ambition of holistic evaluation. In this way, we hope holistic evaluations for language models and for other classes of foundation models will give rise to useful, responsible, and societally beneficial technology.

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This work was made possible by two unique positions that CRFM occupies. First, to develop the framework, build the benchmark, and execute the evaluation, CRFM serves as an interdisciplinary hub for coordinating and uniting the many authors of this work at Stanford University. Second, to acquire the access and resources required to evaluate all the models in this work, CRFM leverages its relationships with the associated foundation model developers.

We further highlight that this work continues a trend documented by Koch et al. (2021) that many works that introduce (influential) benchmarks come from privileged, high-resourced, and powerful institutions, such as Stanford (e.g. Socher et al., 2013; Bowman et al., 2015; Rajpurkar et al., 2016; Srivastava et al., 2021). We actively encourage community-driven approaches to benchmark design, as well as mechanisms to center and drive adoption for benchmarks developed at other institutions. This is of specific importance given how benchmarks embed values and priorities (Ethayarajh & Jurafsky, 2020; Birhane et al., 2022), which should be interrogated, as they are often developed by a small subset of the AI community but go on to orient the work of the broader AI community. For this reason, we are very explicit about our design decisions (e.g. prioritizing user-facing tasks in scenario selection) and directly foreground the limitations of our work with respect to the breadth of considerations in the AI community. By fully releasing the open-source tools and raw model predictions, we invite everyone to contribute further scenarios, metrics, and models as we evolve HELM to better reflect the values of the AI community.

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[88]See https://hai.stanford.edu/foundation-model-issue-brief-series.